Kun Wang

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Education

Princeton University 2025 - Present
Ph.D. in Computer Science Princeton, NJ

- Advisor: Prof. Olga Russakovsky

University of California San Diego 2021 - 2025
Bachelor of Science, Mathematics and Computer Science San Diego, CA

- Research advisors: Profs. Rose Yu and Manmohan Chandraker.
- GPA: 3.98/4.00.

Selected Publications

C=Conference; * equal contribution

[C.1] **Kun Wang**, Xindi Wu, Sanghyuk Chun, Olga Russakovsky, and Esin Tureci. In-Context Learning Can Help Vision-Language Models Overcome Training Priors. *Conference on Neural Information Processing Systems* (*NeurIPS*), 2026.

[C.2] Bohan Lyu, Yucheng Yang, Siqiao Huang, **Kun Wang**, *et al.* MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI. *Conference on Neural Information Processing Systems* (*NeurIPS*), *Datasets and Benchmarks Track*, 2026; oral presentation.

[C.3] Kunal Gupta, Ishit Mehta, **Kun Wang**, Nicholas Chua, Yan Deng, Abhimanyu Krishna, Ravi Ramamoorthi, and Manmohan Chandraker. InteriorAgent: LLM Agent for Interior Design-Aware 3D Layout Generation. *International Conference on 3D Vision* (*3DV*), 2026.

[C.4] **Kun Wang**[*], Sumanth Varambally[*], Duncan Watson-Parris, Yian Ma, and Rose Yu. Discovering Latent Causal Graphs from Spatio-Temporal Data. *International Conference on Machine Learning* (*ICML*), 2025; oral presentation at the NeurIPS Causal Representation Learning Workshop, 2024.

Experience

Amazon May - Aug 2025
Software Development Engineer, AI Seattle, WA

- Helped develop Kiro, AWS's AI software-engineering agent, by designing and optimizing its command-line experience to improve developer productivity.
- Reduced retrieval token usage by 23% across internal web and codebase search while maintaining accuracy; improved SWE-bench performance and drove a 9% increase in internal usage during the first week after deployment.

China Asset Management Co., Ltd. Jul - Sep 2024
Machine Learning Intern Beijing, China

- Developed financial-forecasting models over more than 5,000 stocks, improving prediction accuracy by 15% while aligning technical work with asset-management strategy.
- Built PyTorch time-series pipelines backed by MySQL and deployed scalable services with AWS, Docker, and Kubernetes.

Rose Spatio-Temporal Lab, UC San Diego 2023 - 2025
Undergraduate Researcher with Prof. Rose Yu and Prof. Yian Ma San Diego, CA

- Developed an end-to-end spatio-temporal causal-discovery framework using variational inference, latent causal graphs, and deep causal representation learning.
- Derived identifiability results and optimized GPU-parallel training with PyTorch Lightning, CUDA, and AWS; the work appeared at ICML 2025 and received the HDSI Research Scholarship.

Experience (continued)

Center for Visual Computing, UC San Diego 2023 - 2024
Undergraduate Researcher with Prof. Manmohan Chandraker and Ravi Ramamoorthi San Diego, CA

- Developed a vision-language framework for 3D layout generation and spatial reasoning using LLM-based multi-agent feedback and retrieval-augmented generation.
- Reduced physical and visual errors by 65% versus a diffusion-model baseline; supported by the Qualcomm Innovation Fellowship and published at 3DV 2026.

The Slade Lab, UC San Diego Feb 2022 - May 2023
Undergraduate Research Assistant San Diego, CA

- Built Python and MATLAB probabilistic models to simulate respiratory-aerosol transmission across indoor environments.

Xie Lab, Tsinghua University Jun 2017 - Jul 2021
Summer Research Assistant Beijing, China

- Modeled genetic cell-therapy experiments and analyzed perturbation results to identify potential cancer interventions.

Teaching

Introduction to Computer Vision (CSE 152A), *Teaching Assistant*, UC San Diego 2023

- Supported instruction for CSE 152A, reinforcing core topics including image formation, feature extraction, object detection, and segmentation.
- Provided one-on-one and group tutoring on course material and assignments, adapting explanations to clarify conceptual and programming questions.

Linear Algebra and Differential Equations, *Teaching Assistant*, UC San Diego 2022 - 2023

- Reinforced foundational concepts in linear algebra and differential equations through step-by-step problem-solving guidance.
- Tutored calculus, MATLAB, and mathematical programming, helping students translate mathematical formulations into computational solutions; received 95% positive feedback in teaching evaluations.

Selected Honors and Awards

Princeton University Graduate Fellowship, Princeton University 2025 - 2026
HDSI Undergraduate Scholarship, Halicioglu Data Science Institute 2023
First Place, Taxi Travel-Time Prediction Kaggle Competition 2023

Service and Outreach

Conference Reviewer, *SIGGRAPH Asia, ICML, and NeurIPS* 2024 - 2025

- Reviewed research submissions for SIGGRAPH Asia 2024, ICML 2025, and NeurIPS 2025.

O-STEM: Queer in AI, *Volunteer Organizer* 2024

- Coordinated inclusive UC San Diego panels and discussions, increasing community engagement by 20%; organized a 100-attendee event with Women in STEM on equitable practices in AI.